CSSE 220 – Object-Oriented Software Development
Rose-Hulman Institute of Technology

Worksheet 13

Name (Print):_____ Section:_____

1. **For Loop**.

```
1 public static void countDown(int n) {
2
3     for (int i = __________; i __________; i __________) {
4
5         System.out.println(__________);
6     }
7 }
```

2. **Definition**. Recursion solves a problem by _____.

3. **Recursive version of the for-loop**.

```
1 public static void countDownRec(int n) {
2
3     if (____________________) return;   // base case
4
5     System.out.println(__________);
6
7     countDownRec(_______________);       // recursive call
8 }
```

4. **Essential parts of Recursion**:

   1) _____ (stopping condition)

   2) _____ (reduces the problem)

5. **Identify parts of recursion**:

```
1 factorial(1) = 1                    __________  case
2
3 factorial(n) = n * factorial(n - 1)  __________ case
```

6. **Base case**. What could happen if we didn't have a base case
   Error Type: _____

7. Draw traces

```
1 // sum: n + (n-1) + ... + 1
2 public int tri(int n) {
3   if (n < 1)  return 0;
4   return n + tri(n - 1);
5 }
6 //call from main:
7 tri(3);
```

8. Draw Traces

```
1 public int fact(int n) {
2   if(n == 1)  return 1;
3   return n * fact(n - 1);
4 }
5 //call from main:
6 fact(4);
```